



If the tank begins to overflow, some of the excess water drains out through the bottom holes of the sensor assembly, but sufficient overflow still enters the assembly through the connected pipe to push the float upwards. After a short delay, the reed switch closes due to the magnetic field generated by the float magnet. This pulls pin 13 of the IC3D (74LS08) down to logic 0, corresponding to second line in the truth table.

In this state, the collector voltage of transis-

regulator, LM7805 (IC1), is used, while capacitor C2 helps minimise residual ripple.

To understand the functioning of the relay-control circuit, consider a situation in which the tank is not overflowing and there is insufficient stagnant water inside the sensor assembly to drive the float upwards. In such a state, the float rests on the stopper, keeping the reed switch open.

A 2-input TTL AND gate, IC3D (one-fourth of 74LS08), continuously monitors the open or closed status of the reed switch inside the sensor. Of the four 2-input AND gates (IC3A, IC3B, IC3C, and IC3D) available in IC3, only IC3D is used in this system for monitoring.

Once S1 is switched on, the relay-control electronics connect to the +5V supply line. The output of the AND gate at pin 11 goes high (logic 1) because pin 12 is connected to +5V, and pin 13 also receives +5V through R6, since

the reed switch remains open. As a result, the S input of the S-R latch goes to logic 1.

The R input of the latch momentarily goes low due to the power-on-reset components R3 and C3. Consequently, the S-R latch at pin 3 of IC2 goes to logic 0. Transistor T1 does not conduct, and the collector of T1 rises to 5V.

The Darlington pair (T2 and T3) then receives sufficient base-drive current from the 5V supply through resistors R7, R10, and R12 and enters a highly conductive state. The high emitter current of T2 flows through the base-emitter junction of T3, forcing T3 into saturation. This draws a large current from the +12V supply through the relay coil, energising the relay and closing its normally open contacts. As a result, the pump motor receives the 230V AC supply through the closed contacts and begins lifting water into the tank.

After a short period, the reset

for T1 drops towards 0V, cutting off transistors T2 and T3. As a result, the relay becomes de-energised, switching off the pump motor and stopping the overflow. The stagnant water in the sensor assembly then drains out gradually through the bottom holes.

To operate the pump again, S1 must be switched off, then back on to discharge capacitor C3.

An LED blinker built around IC4 (NE555) is configured as an astable multivibrator, with output pin 3 driving LED3 via resistor R11. When T1 does not conduct, pin 4 of IC4 gets +5V, thereby enabling IC4. As a result, LED3 starts blinking and continues to do so as long as T1 remains non-conductive. This indicates that the pump is running. The combination of capacitor C5 and resistors R8 and R9 determines the blinking frequency. Instead of LED3, a small buzzer can be used to provide an aural indication.

Restarting is possible only when

*Fig. 5: Circuit diagram*